

ORCA: Marine EcOsystem Reasoning with Collaborative Agents

Complete Technical Architecture, Scientific Engines & Implementation Master Guide

Problem Statement ID: SIH26176	Repository: akash20065ray-sys/ORCA
Team Name: CodeCatalysts	Automated Tests: 34/34 Passing (100% Pass Rate)
System Core: Python 3.13 / FastAPI / PostGIS / React 19	Architecture: 8 Domain Agents + Deterministic Engines

1. System Vision & Core Architectural Philosophy

ORCA is an agentic marine decision-support system engineered specifically for India's 7,516 km coastline, encompassing 72 coastal districts and 28+ million active fishermen. Traditional maritime services present severe operational hurdles: IMD weather alerts, INCOIS Potential Fishing Zones (PFZs), ISRO satellite imagery, and Coast Guard security boundaries exist in isolated silos. Fishermen are forced to mentally cross-reference conflicting PDF bulletins, raw GRIB rasters, and VHF radio chatter while operating at sea.

- ORCA resolves this through a **Grounded Multi-Agent Collaborative Architecture** built on three strict non-negotiable principles:
- 1. Zero-Hallucination Guarantee:** The Planning Agent uses an LLM (Anthropic Claude 3.5 Sonnet / Grounded Local Heuristic Engine) solely to comprehend user intent, decompose tasks, and orchestrate specialized agents. *The LLM never fabricates or invents coordinates, wind speeds, wave heights, route geometries, or hazard levels.*
 - 2. Deterministic Scientific Computing:** All domain physics — wave rough-sea risk, Douglas Sea State (0–9), Beaufort Wind Force (0–12), geodesic fairway routing, and thermal front edge detection ($\Delta T/km$) — are calculated by pure mathematical rule engines.
 - 3. Complete Data Provenance:** Every parameter returned carries an auditable citation ledger detailing satellite sensor (e.g. GHRSSST L4 OSTIA, ISRO OCM-3), observation timestamp, retrieval latency, and sensor freshness.

2. End-to-End Multi-Agent Request Pipeline

Every incoming query (via HTTP REST POST /api/chat) traverses a deterministic 8-step pipeline orchestrated by backend/orchestration/orca_orchestrator.py:

Pipeline Stage	Source Module	Functional Responsibility & Outputs
Step 0: Multimodal Ingestion	document_processor.py	Parses uploaded vessel manifests, navigational PDFs, and satellite clips; extracts base64 coordinates.
Step 1: Planning Agent	planning_agent.py	Classifies intent into 10+ categories; geocodes location from 35+ coastal ports; selects required agents.
Step 2: Marine Data Retrieval	marine_data_retrieval_agent.py	Coordinates parallel asynchronous calls across Open-Meteo, NOAA GHRSSST, and INCOIS OMNI buoy telemetry.
Step 3: Ocean Analytics Agent	ocean_analytics_agent.py	Computes SST thermal gradient edge detection ($\Delta T/km$), chlorophyll-a fronts, and thermocline upwelling.
Step 4: Weather Intelligence	weather_intelligence_agent.py	Calculates wind vectors, wind gusts, wave height, swell wave, Douglas Sea State, and Beaufort Force.
Step 5: Alert & Notification	alert_notification_agent.py	Aggregates government bulletins: INCOIS High Wave alerts, IMD Damini lightning strikes, and Port Signals 1–11.

Step 6: Risk Assessment Agent	risk_assessment_agent.py	Executes deterministic multi-factor safety matrix (0–100 score), producing Sail/No-Sail verdicts and causal reasons.
Step 7: Geospatial Analysis	geospatial_analysis_agent.py	Runs PostGIS spatial containment, IMBL proximity alarms (< 12 NM), MPA checks, and obstacle-avoiding routing.
Step 8: Response Synthesis	response_synthesis_agent.py	Synthesizes final grounded report in 12+ native Indian languages with complete Data Provenance ledgers.

3. Mathematical Formulations of the Scientific Engines

A. Marine Risk Assessment Engine (services/risk_engine.py)

The risk engine calculates an integer risk score R in $[0, 100]$ using a multi-factor weighted rule matrix:

$$R = \min(100, R_{\text{wave}} + R_{\text{wind}} + R_{\text{hazard}} + R_{\text{geofence}} + R_{\text{freshness}})$$

- **Wave Risk (R_{wave}):** If $H_s \geq 3.5$ m (Douglas State 5–6), $R_{\text{wave}} = 45$ (CRITICAL). If $2.2 \leq H_s < 3.5$ m, $R_{\text{wave}} = 32$ (HIGH). If $1.5 \leq H_s < 2.2$ m, $R_{\text{wave}} = 18$ (MODERATE). Else $R_{\text{wave}} = 5$ (LOW).
- **Wind Risk (R_{wind}):** If $V_{\text{wind}} \geq 34$ kts or gusts ≥ 42 kts (Beaufort 8+ Gale), $R_{\text{wind}} = 40$ (CRITICAL). If $22 \leq V_{\text{wind}} < 34$ kts, $R_{\text{wind}} = 26$ (HIGH). If $14 \leq V_{\text{wind}} < 22$ kts, $R_{\text{wind}} = 12$ (MODERATE).
- **Hazard Risk (R_{hazard}):** Active Cyclone: +45; IMD Damini Lightning: +25; Port Danger Signal ≥ 3 : +25.
- **Geofencing Risk (R_{geofence}):** IMBL proximity ≤ 12 NM: +30; Marine Sanctuary intersection: +20.
- **Freshness Uncertainty Penalty ($R_{\text{freshness}}$):** If observation latency > 12 hours, an explicit uncertainty margin of +8 points is added.

Decision Verdict: If $R \leq 25$, LOW (Safe to Sail); $26 \leq R \leq 47$, MODERATE (Caution); $48 \leq R \leq 67$, HIGH (Unfavourable); $R \geq 68$, CRITICAL (Prohibited / Evacuate).

B. Navigable Marine Route Engine (services/route_engine.py)

Generates obstacle-avoiding nautical routes strictly preventing land crossing:

- **Route Alpha (Coastal Channel):** Navigates standard offshore fairways with a base safety score of 82.
- **Route Bravo (Deep-Water MPA Bypass):** Arcs seaward via parabolic waypoint expansion to completely avoid shallow reefs and MPAs, achieving higher safety (score 94).
- **Cape Comorin Rounding Protocol:** When routing between the West Coast (Arabian Sea) and East Coast (Bay of Bengal), the engine forces transit waypoints through Cape Comorin TSS Rounding (7.65°N, 77.40°E) and Wadge Bank Deep Transit (7.20°N, 78.50°E), strictly guaranteeing **0% land intersection** across the Indian peninsula.
- **Fuel & Carbon Formulation:**
Fuel Burn Rate (L/NM) = $1.95 \times (V_{\text{knots}} / 12.0)^{1.5}$ | Total Fuel (L) = $D_{\text{NM}} \times \text{Fuel Rate}$
Carbon Emissions = Total Fuel (L) $\times 5.24$ kg CO₂/L

C. Potential Fishing Zone (PFZ) Engine (services/pfz_engine.py)

Combines ground-truthed INCOIS Fish Landing Center (FLC) baselines with live satellite Earth Observation:

- **Thermal Gradient Edge Detection:** Identifies ocean thermal fronts where $\Delta T/\text{km} \geq 0.5^\circ\text{C}/\text{km}$ (confidence +0.06).
- **Chlorophyll-a Upwelling:** Phytoplankton bloom convergence where Chl-a ≥ 1.0 mg/m³ (+0.05). Oligotrophic waters (<0.3 mg/m³) receive a -0.08 penalty.
- **Fleet Range Filtering:** Artisanal motorized crafts are strictly limited to ≤ 15 NM range; mechanized trawlers are routed up to 120 NM.
- **Pelagic Species Association:** Matches Yellowfin Tuna, Skipjack, Mackerel, Sardine, and Squid against depth soundings (35–250 m) and gear recommendations (Ring seine, Gillnet, Longline).

4. Embedded Spatial Index & Geospatial Layer

To achieve sub-5 millisecond response times without requiring external spatial database roundtrips during offline or high-concurrency periods, ORCA embeds a custom **Ray-Casting Point-in-Polygon & Haversine Geodesic Engine** (backend/database/spatial_index.py):

- **Ray-Casting Algorithm:** For any query point (x_0, y_0) and boundary polygon vertices (x_i, y_i) , a horizontal ray is cast to infinity. If the number of edge intersections is odd, the point is interior:
$$\text{Intersect} \iff ((y_i > y_0) \wedge (y_j > y_0)) \wedge \left(x_0 < \frac{(x_j - x_i)(y_0 - y_i)}{y_j - y_i} + x_i\right)$$
- **Pre-Indexed Geofences:** Pre-loaded with official GeoJSON polygons for the **Indian EEZ (200 NM)**, **IMBL Sri Lanka/Pakistan**, and Marine Protected Areas: *Gahirmatha Marine Sanctuary, Gulf of Mannar Marine National Park, Sundarbans Biosphere, and Malvan Sanctuary*.
- **Coastal Port Registry:** 35+ indexed coastal harbors with coordinates, state, and extensive multilingual vernacular aliases.

5. Frontend Architecture: 60 FPS Windy Fluid Canvas & React HUD

ORCA features two production frontend implementations sharing identical API specifications:

- 1. Vanilla Marine-Tech PWA (frontend/):** Zero framework hydration overhead. Built with Vanilla ES6+, CSS3 design tokens, and HTML5 Canvas. Loads in $< 400 \text{ ms}$ on low-spec marine tablets and intermittent 2G cellular networks.
- 2. Modern React 19 + Vite 8 SPA (frontend-react/):** Enterprise component architecture featuring draggable split panels, interactive Leaflet waypoint manipulation, and dynamic voice recognition.

The Windy-Style Fluid Ocean Simulator (windy_animator.js / WindyCanvasEngine.js):

- Renders real-time continuous vector flow fields running locked at **60 FPS** directly on an HTML5 2D Canvas overlaid above Leaflet tiles.
- Animates particle velocity streamlines (Wind), undulating crest wavefront ripples (Swell Waves), and authentic thermal gradient heatmaps.
- **1-Click Instant Unit Conversion:** Live switching between kt / km/h / m/s / mph, m / ft, and $^{\circ}\text{C}$ / $^{\circ}\text{F}$ with an interactive bottom scale bar and live cursor-tracking probe.

6. GMDSS Emergency Distress Mayday System

Located at backend/api/emergency.py, the distress module provides IMO-compliant search and rescue coordination:

- **Coast Guard MRCC Integration:** Automatically geolocates and routes to the nearest Maritime Rescue Coordination Centre: *MRCC Kochi (+91-484-2216444), MRCC Mumbai (+91-22-24388065), MRCC Chennai (+91-44-23460405), or MRCC Port Blair (+91-3192-233633)*.
- **AIS Traffic Proximity:** Scans simulated coastal AIS traffic along shipping lanes, identifying closest commercial feeders and Coast Guard Fast Patrol Vessels (e.g. *ICGS Samar* on VHF Ch 16 / DSC Ch 70).
- **SAR Intercept ETA:** Calculates Coast Guard Fast Interceptor Craft (FIC) transit time based on 35-knot operational speed.
- **IMO Standard Mayday Script:** Automatically formats standard voice script: MAYDAY, MAYDAY, MAYDAY. THIS IS IND-KL-07-ORCA....
- **Cryptographic Dispatch Token:** Issues SHA-256 tokens (e.g. GMDSS-MRCC-KOC-A4F82B91) for audit tracking, stand-down protocol, and downloadable PDF receipts.

7. Comprehensive Technology Stack & Libraries

Component Layer	Technology & Version	Architectural Justification & Role
Backend Framework	Python 3.13 / FastAPI / Uvicorn	Asynchronous, high-throughput microsecond API execution.
Data Validation	Pydantic v2	Strict type-safe schemas for observations, routes, and telemetry.
AI Reasoning	Anthropic Claude 3.5 Sonnet	Intent understanding and structured multi-agent coordination.

Fallback AI	Deterministic Heuristic Engine	Zero-latency (<0.5ms) local fallback guarantee when offline.
Spatial Database	Supabase (PostgreSQL + PostGIS)	Cloud spatial persistence with R-Tree indexing.
Embedded Spatial	Custom Ray-Casting Engine	Zero-dependency sub-millisecond point-in-polygon queries.
Frontend (React)	React 19 / Vite 8 / Lucide React	Modern reactive SPA with interactive map stage and HUD.
Frontend (PWA)	Vanilla ES6+ / HTML5 Canvas	Locked 60 FPS fluid simulation without framework overhead.
Mapping	Leaflet.js 1.9.4	Interactive nautical charts, GIS overlays, and draggable waypoints.
Voice & Speech	Web Speech API (STT & TTS)	Hands-free vernacular audio interaction across 12+ Indic languages.
Testing Suite	Pytest 9.1 / AnyIO	34 automated unit, connector, and SIH scenario tests.

8. Automated Test Suite & Quality Verification

The system underwent rigorous automated verification using pytest. All 34 tests pass with 100% success rate:

- **test_chat_intents.py (6 tests):** Greetings, vernacular phrases, out-of-domain guardrails, Kallakkadal oceanography, port danger signals 1–11, and telemetry retention.
- **test_connectors.py (4 tests):** Open-Meteo, NOAA/ISRO SST & Chlorophyll, INCOIS bulletins, and GIS boundaries.
- **test_new_backend_apis.py (9 tests):** Emergency SOS broadcast, MRCC registry, weather forecast timelines, in-situ buoys, active alerts, safety endpoint, PFZ diagnostics, route calculation, and session persistence.
- **test_risk_engine.py (3 tests):** Calm conditions safety score, rough sea/cyclonic storm risk, and stale data uncertainty penalty (+8 pts).
- **test_route_engine.py (1 test):** Marine route calculation, fairway waypoint generation, and distance math.
- **test_sih_scenarios.py (10 tests):** All 10 official Smart India Hackathon problem statement challenge queries.
- **test_ui_endpoints.py (1 test):** SPA root mounting, static assets, dynamic steerage instructions, and GIS map layers.